Exact referee relaxations for Levine's two-player hat problem

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Abstract

We prove $V_2 \leqslant 2383/6720 \approx 0.354613$ for the two-player Levine hat problem, improving the previous bounds $81/224 \approx 0.361607$ and $3795217/10582656 \approx 0.358626$. The proof works with the common-row referee relaxations H_d of [W], which satisfy $V_2 \leqslant \max$ H_d for every d.

First we solve the d = 4 relaxation exactly: $\max H_4 = 57/160 = 0.35625$, attained by 11 symmetry classes (250,880 labelled partitions). Then a hierarchical search in dimension 5 caps each of the 31 hyperplanes using complete lists of near-optimal d = 4 partitions, and scores exactly the few complete partitions that survive. The bound closes 60% of the gap between 81/224 and the conjectured value 7/20.

Simulated annealing suggests $\max H_5 \approx 1401/3968 \approx 0.353075$ and $\max H_6 \geqslant 45/128 \approx 0.351563$, consistent with the relaxations converging to 7/20. All results are computer-assisted and not yet peer reviewed; the code, logs and independent checks accompany this note.

1. Introduction and results

Two players each wear an infinite stack of independent fair black or white hats. Each sees only the other's stack and points to one hat in their own; they win if both chosen hats are black. Let V_2 be the supremum of the winning probability over all strategies (equivalently, over finite stacks).

The best strategies win with probability 7/20, and Bouquet et al. conjecture $V_2 = 7/20$ [BCCZW]. Upper bounds have moved slowly. The trivial bound is 3/8; a computer-assisted argument of Freiling et al. gives 81/224 [BFG+]; a Fourier approach gives 0.37193 without large searches [HT]. Our earlier write-up [W] introduced a common-row referee and reached 3795217/10582656.

Source	Upper bound on V_2	Decimal
Trivial	3/8	0.375000
Heilman–Tamuz [HT]	—	0.371930
Freiling et al. [BFG+]	81/224	0.361607

Source	Upper bound on V_2	Decimal
[W]	3795217/10582656	0.358626
Theorem 1 (this note)	57/160	0.356250
Theorem 2 (this note)	2383/6720	0.354613

Theorem 1. max_P $H_4(P) = 57/160$. Hence $V_2 \leqslant 57/160$.

Theorem 2. max_P $H_5(P) \leqslant 4766/13440 = 2383/6720$. Hence $V_2 \leqslant 2383/6720 \approx 0.354613$.

Theorem 1 is the exact value of the d = 4 relaxation, so no argument based on H_4 alone can do better. Theorem 2 needs a genuinely five-dimensional search. Both are computer-assisted, and both rely on the referee reduction and the averaging inequalities of [W] (Section 2), which we checked by hand.

We also report evidence about the limits of the method (Section 7). The best d = 5 partition we found has $H_5 = 1401/3968 \approx 0.353075$, and the best d = 6 partition we found has $H_6 = 45/128 \approx 0.351563$. The relaxation values fall by roughly half each dimension, consistent with convergence to 7/20, but this is extrapolation.

2. The referee relaxation

Fix $d \geqslant 1$. A label is a pair (a, b) with $a \in F_2^d \setminus \{0\}$ and $b \in F_2$; there are $m_d = 2(2^d - 1)$ labels, and label (a, b) covers the point $x \in F_2^d$ when $a \cdot x \oplus b = 1$. Equivalently, labels are the affine hyperplanes of AG(d, 2) and a label covers its own points. For a partition P of the labels, keep each block independently with probability 1/2, let U be the union of kept blocks, and set

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H_d(P) = \frac{1}{m_d}\, \mathbb{E} \max_{x \in \mathbb{F}_2^d} \#\{(a,b) \in
U : a \cdot x \oplus b = 1\}.
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Lemma 1 (referee, [W]). $V_2 \leqslant$ max_P H_d(P) for every d.

Sketch. Draw a hidden label J = (a, b) uniformly, independent uniform rows $R_i \in F_2^d$, and Bob′s fair string Z. Alice′s bits $a \cdot R_i \oplus b$ are independent and fair for each fixed J, so the game is unchanged. Reveal R to both players and J to Bob; extra information only helps. For fixed R, Bob′s choice partitions the labels by the coordinate he selects, and the selected coordinates of Z keep blocks independently with probability 1/2. Alice then chooses a row, and allowing every $x \in F_2^d$ only helps her.

We write all d = 4 and d = 5 values in units of 1/13440, so 81/224 = 4860, 57/160 = 4788, 227/640 = 4767, 2383/6720 = 4766 and 7/20 = 4704.

Lemma 2 (averaging, [W]). For a three-dimensional subspace W of the coefficient space (a *frame*), let F_W be the same objective restricted to the 14 labels with a ∈ W. Every label lies in the same number of frames, and every frame in the same number of hyperplanes T, so taking maxima separately can only increase the value:

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H_d(P) \le \frac{1}{N_d} \sum_{W} F_W(P|_W), \qquad H_d(P) \le \frac{1}{2^d - 1} \sum_{T} H_{d-1}(P|_T),
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where N_d is the number of frames (15 for d = 4, 155 for d = 5). The restrictions keep the fair independent block law because distinct blocks stay distinct.

Lemma 3 (symmetry). H_d is invariant under G_d = AGL(d, 2) × ⟨σ⟩, acting on labels by (a, b) ↦ (Ma, b ⊕ a·c ⊕ s). Linear maps and translations relabel the points; the complement swap σ: b ↦ b ⊕ 1 preserves H_d because each point is covered by exactly half the labels and U has the same law as its complement, so E max count(σU) = m_d/2 − E min count(U) = E max count(U). G_d has order $2 \cdot 2^d \cdot |\mathrm{GL}(d, 2)|$: 2,688 for d = 3, 645,120 for d = 4.

3. Frames and the d = 3 local table

The frame objective is the d = 3 relaxation, and its maximum is 324/896 = 81/224. We scored all Bell(14) = 190,899,322 partitions of the 14 frame labels exactly, in 42 s on two cores. Each score is an integer $N = 2^{14} \cdot E\max$ (so $F = N / (14 \cdot 2^{14})$), computed by a Gray-code walk over all subsets of blocks.

Frame score (×1/896)	Partitions	Orbits under G_3
324	3,920	5
322	16,128	8
321	1,344	1
320	159,712	79
319	75,264	31
318	545,104	223
317.75	2,256	5
317.5	11,648	6
317	564,928	227

The first five counts agree with [W]. No score lies strictly between the listed levels, and every one of the 1,380,304 listed partitions keeps its score under all 2,688 elements of G_3.

For a partition P of the d = 4 labels, write R(P) = Σ_W (324 − 896 F_W) for its *frame loss* (exact, not rounded). Since 15 · 896 = 13440, Lemma 2 becomes the frame bound

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H_4(P) \le \frac{4860 - R(P)}{13440}.
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Alongside the full table we store, for every prefix of a restricted-growth string, the largest score over all completions. A frame whose first j labels are placed is then bounded by one table lookup. One lemma lets a single table serve every frame in every dimension.

Lemma 4 (sorted affine subspaces). List an affine subspace of F_2^n in increasing numerical order as $o_1 < o_2 < \dots$. Then $o_{r+1} = o_1 \oplus \Sigma_k r_k b_k$, where r_k are the binary digits of r and b_k is the reduced row-echelon basis ordered by leading bit.

Proof. Take o_1 reduced against the basis. Two elements differ first at the highest pivot where their basis combinations differ, so numerical order is the binary order of the combinations, and o_1 is the minimum. ∎

Consequently, if the labels outside a fixed subspace are placed in increasing order of a, any frame or hyperplane receives them in an order that a linear isomorphism carries to the canonical order.

4. Exact solution of the d = 4 relaxation

The maximum of H_4 is 57/160, reached by 11 orbits of partitions. We find it by branch-and-bound against a threshold L: we show every partition satisfies $H_4 \leqslant \max(T_L, \text{best exact leaf})$, where T_L = (4860 − L)/13440.

Search. Every partition has a frame of maximal score. By Lemma 3 we may move it to the frame of labels 0–13 (direction 8) and make its restriction an orbit representative R. All other frames then score at most N_R, so if 15 times R′s loss is at least L the frame bound already gives $H_4 \leqslant T_L$. Only representatives of smaller loss are searched, and the program checks this list against the full histogram.

The 16 remaining labels are placed in increasing order as a restricted-growth string. By Lemma 4, each of the other 14 frames sees its 6 shared labels first and its 8 new labels in canonical order, so the prefix table bounds it, clamped at N_R. A node is pruned when the 15 frame bounds sum to at most 256(4860 − L). Surviving complete partitions are scored exactly over all 2^K block subsets.

Symmetry. The 16 elements of G_4 that fix every frame label act regularly on the 16 outside labels and fix the root′s blocks. So we may require label 14 to join the lowest-numbered root block used by any outside label. We also apply a lex-leader test, to depth 8, under every lift of R′s stabilizer in G_3. Both reductions keep at least one representative of every orbit (Section 6).

L	Root orbits	Nodes	Surviving leaves	Best leaf (×1/13440)	Time, 2 cores
40	13	1.80×10^{7}	6,728	4788	0.4 s
56	14	3.44×10^{8}	530,680	4788	8 s
72	93	6.79×10^{9}	16,671,664	4788	156 s
80	124	3.30×10^{10}	89,969,576	4788	12 min
88	124	2.76×10^{10}	93,598,369	4788	21 min
96	352	6.20×10^{10}	120,574,288	4788	23 min
104	358	3.85×10^{11}	875,667,299	4788	1.9 h

Runs from L = 88 use the kernel reduction, and from L = 96 the lex-leader test too, so leaf counts are of representatives.

Proof of Theorem 1. At L = 72 the threshold T_{72} equals 4788/13440 = 57/160 and no leaf exceeds it. Hence max $H_4 \leqslant 57/160$, and the leaves attaining it give equality. ∎

The optima form 11 orbits, 250,880 labelled partitions. Every optimum has 12 frames at 324/896, frame loss 36 or 40, and 5 or 6 blocks. One optimum with 5 blocks, writing a as a 4-bit vector:

- {(0001,0), (0010,0), (0100,0), (0111,0), (1001,0), (1010,0), (1100,0), (1111,0)}
- {(0001,1), (0010,1), (0101,0), (0110,0), (1000,1), (1001,1), (1010,1), (1011,0), (1101,0), (1110,0)}
- {(0011,0), (1011,1)}
- {(0011,1), (1000,0)}
- {(0100,1), (0101,1), (0110,1), (0111,1), (1100,1), (1101,1), (1110,1), (1111,1)}

Near-optimal partitions. The L = 104 run lists every partition with $H_4 > 4760/13440$. They take only nine values:

H_4 (×1/13440)	H_4	Orbits	Labelled partitions
4788	57/160	11	250,880
4781	683/1920	48	1,320,480
4774	341/960	57	2,856,000
4770.5	1363/3840	15	1,854,720
4768.75	545/1536	13	2,419,200
4767.875	5449/15360	11	685,440
4767	227/640	69	8,534,400

H_4 (×1/13440)	H_4	Orbits	Labelled partitions
4765.25	2723/7680	34	2,499,840
4763.5	1361/3840	33	3,427,200

In total 291 orbits and 23,848,160 labelled partitions. The values 341/960 and 1363/3840 also appear in [W] as maxima of its $A_0 = 11$ and $A_0 = 10$ families.

Envelope. The same runs give g(r), the largest H_4 among partitions with frame loss at least r. It falls fast once frame loss passes 40: g = 4788 up to r = 40, 4781 at 42, 4774 for 44–48, 4763.5 at 50, 4760 for 52–78, 4746 at 80–82, 4739 at 84 and 4732 for 86–103. Beyond the enumerated range only the frame bound 4860 − r applies, so after L = 104 the usable floor is 4756 for r ⩾ 80.

5. A hierarchical search in dimension 5

Theorem 2 comes from a search over partitions of the 62 labels of F_2^5 that bounds H_5 by the hyperplane average of Lemma 2, $H_5 \leqslant (1/31)\ \Sigma_T\ H_4(P|_T)$. The point is that each hyperplane is a copy of the d = 4 problem we have just solved, so its value can be capped long before it is complete.

Root. For a target τ, every partition has a hyperplane of maximal H_4. If that maximum is at most τ, the average is at most τ. Otherwise, by Lemma 3 in dimension 5, we move it to T_{16} (the labels with a < 16, which are exactly the d = 4 labels) and make its restriction an orbit representative. So the roots are the d = 4 orbits with $H_4 > \tau$, listed completely by Section 4 whenever $T_L < \tau$. Every other hyperplane then has H_4 at most the root value h_0.

Caps. The 32 outside labels are placed in increasing order. Each non-root hyperplane shares one frame (14 labels) with T_{16} and gains 16 outside labels; its other 14 frames each have 6 shared and 8 new labels. A hyperplane's value is bounded by the minimum of:

1. the root value h_0;
2. its frame bound, from the prefix table of Section 3 for each of its frames;
3. the envelope g(r), with r the frame loss already forced by those prefixes;
4. a **level cap**. By Lemma 4 a linear isomorphism sends the hyperplane to the d = 4 labels with its outside labels at positions 14–29 in placement order. Its partial assignment is then a prefix of a restricted-growth string. A binary search in the sorted list of all 23.8 million labelled partitions above the floor, plus a range-minimum query, gives the best level any of them reaches from this prefix. If none extends it, the cap is the floor itself;
5. its exact H_4 once all 30 of its labels are placed.

A node is pruned when h_0 plus the 30 caps is at most 31τ, or when a completed hyperplane beats h_0 (that partition belongs to another root). Complete partitions that survive are scored exactly.

Symmetry. 32 elements of G_5 fix every root label and act regularly on the outside labels, so label 30 may be required to have minimal type as in Section 4. A lex-leader test to depth 12 then uses every lift of the root's stabilizer, between 512 and 18,432 elements per root.

Proof of Theorem 2. The run at $\tau = 4766$ uses the $L = 104$ lists (floor 4760) and the 224 orbits with $H_4 \geqslant 4767$; no $d = 4$ value lies strictly between 4765.25 and 4767. It split into 19,821 tasks run in 133 resumable chunks and visited 3.12×10^{10} nodes in 9.9 hours of computing. In the first 750 tasks, whose roots lie nearest the best known partition, 2,237 complete partitions survived the caps. Each was scored exactly, and the largest is $1401/3968 \approx 0.353075$, the best value known. Hence $\max H_5 \leqslant 4766/13440 = 2383/6720$. ∎

Target τ (×1/13440)	Bound	Roots	$d = 4$ data	Nodes	Time
4787	0.356176	11	frame bounds only	1.91×10^9	208 s
4781	0.355729	11	$L = 80$ lists	2.15×10^8	220 s
4777	0.355432	59	$L = 88$ lists	1.10×10^8	294 s
4774	0.355208	59	$L = 96$ lists	2.09×10^8	289 s
4771	0.354985	131	$L = 96$ lists	3.46×10^9	54 min
4769	0.354836	131	$L = 104$ lists	7.94×10^9	114 min
4767	0.354688	155	$L = 104$ lists	1.78×10^{10}	5.2 h (40 chunks)
4766	0.354613	224	$L = 104$ lists	3.12×10^{10}	9.9 h (133 chunks)

Down to $\tau = 4767$ no run produced a surviving partition; at $\tau = 4766$ the 2,237 survivors were all scored exactly. The floor drives the cost: with the $L = 88$ lists the target 4774 had covered 7% of its tasks after 13 minutes, while with the $L = 96$ lists it finished in under 5. Each background job here was limited to 2 hours, so the $\tau = 4767$ and $\tau = 4766$ runs were split into chunks (smaller ones where tasks were slow) and resumed across many jobs and machine restarts; every chunk log records identical settings.

Profiling at $\tau = 4777$ shows where the work is. Most nodes sit between labels 39 and 43, and the largest single cut comes at label 45, when the first non-root hyperplane completes: 1.3 million nodes enter, about 1,800 survive.

6. Verification

Every program was written from the definitions alone, and each search was checked against independent code or a slower variant. The table lists what was tested.

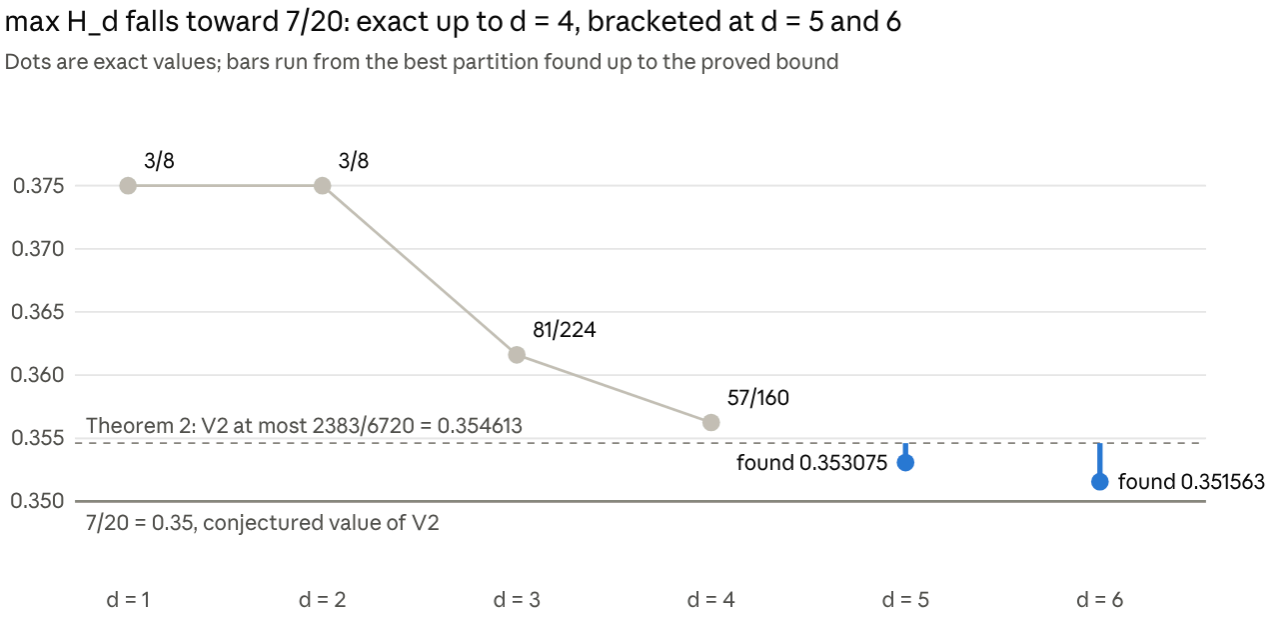
What	Check	Result
Local table	300 partitions rescored by direct subset summation in Python with exact fractions	0 mismatches
Local table	Level counts against [W]	identical for all five levels
Symmetry, d = 3	All 2,688 elements of G_3 on all 1,380,304 partitions scoring ⩾ 317/896	score always preserved
Witness of [W]	Frame scores, average and frame loss	10 × 324, 1 × 320, 4 × 318; 151/420; 28 (exact H_4 = 45/128)
d = 4 bookkeeping	Incremental frame indices against direct recomputation, all 530,680 leaves at L = 56	0 mismatches
d = 4 completeness	Partitions of small frame loss found by an unrelated Python hill-climb, mapped into canonical form by an explicit element of G_4	15/15 found (root 324, L = 56); 8/8 found (root 322, L = 72)
d = 4 symmetry reductions	Envelope at every exact frame-loss value, and orbits of optimal leaves, with and without reductions at L = 72	identical; 11 orbits each time
d = 4 exact scores	400 best leaves and all 832 optimal leaves rescored in Python	0 mismatches
Frame averaging	Lemma 2 on 60 random partitions	0 violations
Orbit lists	Distinct images against stabilizer-based orbit size, all 291 orbits	equal
Hyperplane maps, d = 5	H_4 of each hyperplane through its d = 4 map against direct evaluation, random partitions	0 mismatches
Symmetry, d = 5	Lifted root stabilizers on 1,000 random completions	H_5 and root partition preserved

What	Check	Result
d = 5 caps	70 full partitions (random, hill-climbed, and lifts with up to 16 hyperplanes ⩾ 4781) replayed through the search: every cap ⩾ that hyperplane's final H_4, total ⩾ $31 \cdot H_5$, at every depth	0 violations in 2,310 node checks
d = 5 variants	Range-minimum and nested-list versions at τ = 4780; kernel-only and lex-leader versions at τ = 4777	identical node counts at 4780; no survivors in either at 4777
Chunked runs	Every task covered exactly once; every survivor scored exactly, none above τ	complete for L = 96, L = 104, τ = 4769, τ = 4767 and τ = 4766
Survivors at τ = 4766	Each chunk's best survivor rescored independently in numpy with exact integers; every survivor's exact H_5 checked against its own capped hyperplane average	0 mismatches (best 1401/3968); 0 failures over 2,237 survivors

Two inputs are not verified by code. The referee reduction (Lemma 1) and the averaging inequalities (Lemma 2) are taken from [W]; we checked their proofs by hand. All code ran on one machine, and nobody else has reviewed it.

7. Limits of the method and outlook

The method of Section 5 is close to its own ceiling, while the relaxations themselves appear to keep falling toward 7/20.



Exact values: brute force (d = 1, 2), Section 3 (d = 3), Theorem 1 (d = 4). Lower ends at d = 5, 6: simulated annealing. Upper ends: Theorem 2, using max H6 ⩽ max H5.

Simulated annealing with exact scoring finds the d = 4 optimum on its first restart. At d = 5 the best of 11,360 restarts over 20 minutes is 1401/3968 ≈ 0.353075, found immediately and never beaten. At d = 6 the best of 6,493 restarts is 45/128 ≈ 0.351563. The drops between dimensions (0.0134, 0.0054, 0.0032, then at least 0.0015) roughly halve each time, consistent with a limit of 7/20, though this is extrapolation.

The averaging ceiling. The best d = 5 partition has hyperplane values 8 × 4788, 20 × 4781, 1 × 4606 and 2 × 4588.5, averaging 4764.74, about 19 units above its true H_5 of 4745.32. So the hyperplane average that every bound in Section 5 rests on has its maximum between 4764.74 and 4767. At τ = 4766 the search already scores a few thousand complete partitions exactly; below about 0.35452 it would have to score vast numbers, so Theorem 2 is within 10^{-4} of what this method can give.

Coupled relaxations. The averaging loss comes from letting each hyperplane choose its own row. Since every label lies in 15 hyperplanes, H_5 can also be bounded by averaging over pairs or triples of hyperplanes that must share a row: each such group is weighted so the counts sum to 15 · c(x), and maxima are taken group by group. On the best d = 5 partition these bounds tighten quickly:

Bound on H_5 for the best partition	Value (×1/13440)
Hyperplanes, each with its own row (Lemma 2)	4764.74
Pairs of hyperplanes sharing a row (465 pairs)	4756.79
Triples sharing a row (4,495 triples)	4749.51

Bound on H_5 for the best partition	Value (×1/13440)
True H_5	4745.32

A search built on triples could in principle approach the true d = 5 value. The difficulty is timing: a group can be scored exactly only once all its hyperplanes are complete, which happens in the last eight or so outside labels, while most of the search cost lies earlier.

Open directions.

1. A d = 5 bound that does not let each hyperplane choose its own row would aim at the true value, about 0.3531.
2. Dimension 6 by the same hierarchy needs complete lists of near-optimal d = 5 partitions, beyond brute force.
3. Does max H_d converge to V_2? If so, and V_2 = 7/20, the referee family is a route to the conjecture.
4. On the lower side, no strategy is known to beat 7/20, and no finite stack attains it [BCCZW].

8. Reproducibility

All computations ran on one 2-core x86-64 machine with 8 GB of memory, in about 28 hours of wall-clock time in total. The code is C++20 with SSE4.1, plus Python 3 for the independent checks; it is in the archive levine_d5.tar.gz with its README and every run log.

Program	Role
local3	Scores all Bell(14) frame partitions; writes the full table (764 MB) and prefix maxima
rootclass	Orbits of frame patterns under G_3; checks score invariance
bb4, bb4s, bb4x, bb4y	d = 4 branch-and-bound: plain, kernel symmetry, lex-leader, streamed leaves
orbit4, expand4	Orbit representatives under G_4; all labelled images, sorted
bb5, bb5L, bb5S, bb5R	d = 5 search: frame caps, level caps, lex-leader, range-minimum level caps
run_chunks.sh, run_d5.sh	Resumable chunked runs

Program	Role
merge_chunks.py, check_d5.py	Coverage checks over chunks; any survivor must be scored exactly at or below τ
pyref.py, verify.py, completeness.py, spotcheck_local.py	Independent Python checks (Section 6)
anneal, anneal6	Simulated annealing for max H_d, d = 4, 5, 6

To reproduce Theorem 2: build the frame table, run the d = 4 search at L = 104 in chunks, reduce and expand the orbits above 4760, then run the d = 5 search at τ = 4766 with those lists and the 224 roots above 4766 and check the chunks. The README lists the exact commands.

References

- **[BCCZW]** C. Bouquet, S. Chikhi, T. Charles, Y. Zhou, E. Wang. An analytical framework for the Levine hats problem: new strategies, bounds and generalizations. arXiv:2508.01737, 2025.
- **[BFG+]** On Levine's notorious hat puzzle, arXiv:1407.4711; an earlier version is J. Kariv, C. Van Alten, D. Yeroshkin, *Computing optimal strategies for a cooperative hat game*. Cited as [BFG+22] in [HT] for the bounds 7/20 and 81/224, the latter credited to Freiling et al.
- **[HT]** S. Heilman, O. Tamuz. A Fourier approach to Levine's hat puzzle. arXiv:2503.09042, 2025.
- **[W]** A compact computer-assisted improvement for Levine's two-player game. Write-up by the author of this note, 2 November 2026. Included with the code as paper/write-up-2026-11-02.md.